Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

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Abstract

Climate-driven shifts in stock productivity reach fishing fleets unevenly when species portfolios differ across segments and the sector-level allocation of catch rights cannot adjust to relative productivity changes. We quantify this distributional channel for the small pelagic fishery of Central-South Chile (anchoveta, sardine, jack mackerel), managed under a fixed industrial–artisanal allocation. We couple a Bayesian state-space model of stock dynamics, fit to 2000–2024 official assessment data, with a vessel-level trip equation; climate enters through an indirect biomass channel and a direct weather channel. Under a six-model CMIP6 ensemble, projected fleet-level effort falls by 16–18 percent for the artisanal fleet but only 0.8–1.0 percent for the industrial fleet—an asymmetry of roughly seventeen to one driven by differential portfolio exposure and weather sensitivity. The findings imply that climate-adaptation policy should revisit the sector-level allocation of catch rights, allowing it to respond to relative productivity shifts.

Keywords: Bayesian state-space model; Bioeconomic projection; Chilean small pelagic fishery; Climate change; Fishing effort; Fleet heterogeneity; Quota allocation

JEL Codes: Q22, Q54, Q57, Q58

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Introduction

Climate change is reshaping the productivity of marine fisheries, but the incidence of this reshaping across fleet segments is rarely uniform. Quota-allocation regimes that fix the industrial–artisanal split of the catch were typically designed under stationary-climate assumptions, so a fleet-asymmetric productivity shock acts as a de facto reallocation of harvest rights. Whether such regimes should anticipate this reallocation depends on the projected magnitude and asymmetry of fleet-level effort responses under credible climate scenarios—a quantity that the empirical literature has rarely delivered jointly for multi-species fisheries managed under sector-asymmetric allocation regimes.

We provide that quantity for the small pelagic fishery (SPF) of Central-South Chile (from Valparaíso to Los Lagos), part of the country's largest fishery—approximately 86% of national fish landings by industrial and artisanal sectors in 2019 (SUBPESCA 2020)—and the segment in which all three target species, anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*; locally, sardina común), and jack mackerel (*Trachurus murphyi*; locally, jurel), coexist in non-negligible shares of both fleet portfolios. These three species share habitat and are coupled through trophic and market linkages (Alheit and Niquen 2004; Arancibia et al. 2019; Dresdner et al. 2013). The fishery is managed through a sector-specific quota architecture: a fixed industrial-versus-artisanal allocation splits the Total Allowable Catch between the two segments, while within-sector distribution operates through Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) for the industrial fleet and the Artisanal Extraction Regime (*Régimen Artesanal de Extracción*, RAE) for the artisanal fleet, with limited cross-sector transferability. Because the two segments target different species portfolios, a stock-asymmetric climate shock translates directly into a fleet-asymmetric effort response. Cross-country evidence indicates that Chile is projected to lose part of its maximum catch potential under climate change: Cheung et al. (2010) classify Chile among a group of countries with moderate projected declines (6–13% by 2055), including the contiguous United States, China and Brazil; the available climate-fisheries economics literature is patchy globally and concentrated in a few well-studied regions (Sumaila et al. 2011). To our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.[1]

Our bioeconomic framework combines two empirical components. First, a Bayesian state-space model of multi-species stock dynamics — fitted to 2000–2024 official assessments from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) and the South Pacific Regional Fisheries Management Organisation (SPRFMO) — estimates, for each species, two semi-elasticities of productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies. We call these parameters *climate shifters*. Because they enter the biological law of motion as coefficients on exogenous climate forcings, they retain a structural interpretation when applied to climate states outside the 2000–2024 sample. Second, a vessel-level negative binomial trip equation — estimated separately for the industrial and artisanal fleets, with year fixed effects — maps prices, allocated harvest, weather, and regulatory closures into annual fishing effort. Climate reaches fleet-level effort through two channels: an *indirect biomass channel* via the climate shifters, and a *direct weather channel* via vessel-specific exposure to severe winds. We evaluate the framework against a six-model ensemble from the Coupled Model Intercomparison Project Phase 6 (CMIP6) under two Shared Socioeconomic Pathway scenarios, SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions), at mid-century (2041–2060) and end-of-century (2081–2100).

We obtain three sets of results. First, the artisanal segment absorbs most of the projected climate impact: fleet-level effort falls by 16–18% for artisanal vessels but only 0.8–1.0% for industrial vessels under SSP5-8.5, an asymmetry of roughly seventeen to one. Second, this asymmetry is driven by differential portfolio exposure combined with limited cross-sector transferability: the artisanal fleet is concentrated in anchoveta and sardine, whose climate shifters are large and well-identified, while 95% of the industrial fleet's portfolio sits in jack mackerel, whose Centro-Sur climate shifter the data cannot identify. Third, the within-artisanal heterogeneity is itself first-order: the dominant *lancha motor* segment contracts by −19.6%, the smaller *bote motor* segment by −10.0%.

The remainder of the paper is organized as follows. The next section describes the economic and institutional setting of the Chilean SPF. We then present the empirical framework — data, stock-dynamics model, trip equation, and projection design — followed by results on identified climate shifters, comparative-statics projections of stock productivity, and projections of fleet-level effort.

[1] The only empirical study of Chilean fisher behavior that we are aware of is Peña-Torres et al. (2017), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

The paper closes with a discussion of policy implications and limitations, and brief concluding remarks.

The small pelagic fishery in Chile

The fishery is divided into two latitudinal zones with distinct species compositions. The northern zone (from Arica y Parinacota to Coquimbo) is dominated by anchoveta and jack mackerel, while sardine is effectively absent. In contrast, the Central-South zone (from Valparaíso to Los Lagos) includes all three species in non-negligible shares within both artisanal and industrial landings. Because this region supports a genuinely multi-species fishery operating under a fixed sector-level allocation regime, it provides the relevant setting for evaluating the distributional consequences of climate-induced shifts in stock productivity, and is therefore the focus of this paper. The principal ports servicing this zone include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. The three species are harvested primarily by purse-seine fleets that differ in vessel size, operating range, and species composition.

The historical evolution of the fishery illustrates the portfolio mechanism developed in the remainder of the paper. The jack mackerel fishery was initially concentrated in northern Chile but shifted toward the Central-South zone during the mid-1980s (Peña-Torres et al. 2017), where fishing grounds have traditionally operated within 50 nautical miles of the coast. Between 1987 and 2004, approximately 85% of jack mackerel landings were directed to the reduction industry (Peña-Torres et al. 2017), reflecting the importance of fishmeal and fish-oil export markets in determining the value of the industrial portfolio. Sardine and anchoveta, by contrast, are jointly harvested using the same gear and fishing grounds. The two species share habitat and fishing technology, making species differentiation at the point of capture effectively impossible (Dresdner et al. 2013). As a result, the binding constraint on the species mix of landings is the quota regime, not biology or technology.

The status of the three stocks has evolved substantially over time. Central-South anchoveta was classified as collapsed through 2018, reclassified as overexploited in 2019, and has been harvested within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within

MSY reference levels, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel remained overexploited until 2018 and has since been harvested within MSY limits.

Biological closures (*vedas*) constitute a central management instrument for sardine and anchoveta. In the Central-South zone, reproductive closures extend from July to October in Araucanía–Los Lagos and from August to October in Valparaíso–Biobío, while recruitment closures occur from January through February across all regions. In total, these regulations generate 151 closed fishing days per year in Valparaíso–Biobío and 182 days in Araucanía–Los Lagos. Jack mackerel is not subject to seasonal biological closures.

The Chilean SPF is managed through an annual Total Allowable Catch (TAC; *Cuota Global*), further subdivided across regions and seasons, with unused quotas reassignable within the fishing year and small fractions reserved for research and contingency. Anchoveta and sardine operate under a mixed-species quota framework in which each species has an independent quota but substitution between them is permitted.

The partition of each TAC between the industrial and artisanal sectors is fixed by statute. The pre-reform partition governed the de jure allocation during the estimation window (2013–2024); a 2025 reform (Congreso Nacional de Chile 2025) fixes a new partition in force until 2040 (Table 1). Under the pre-reform regime, the artisanal sector held 78% of the Central-South anchoveta and sardine TACs and 10% of the jack mackerel TAC; the 2025 reform raises the anchoveta and sardine artisanal shares to 90% across Valparaíso–Los Lagos, and the jack mackerel artisanal share to 30% in Valparaíso–Los Ríos and 15% in Los Lagos. The de jure partition admits a single administratively rationed inter-sectoral re-allocation channel, subject to ex-ante authorisation by the Subsecretaría de Pesca y Acuicultura (SUBPESCA). Quota control reports from the Servicio Nacional de Pesca y Acuicultura (SERNAPESCA) for 2013–2024 document that the industrial sector cedes a persistently large share of its assigned anchoveta and sardine TACs to artisanal cessionaries — from 12–72% over 2013–2017, tightening to 85–99% every year in 2018–2024 (Online Appendix, Table H.3) — while jack mackerel cessions are an order of magnitude smaller and shift direction across years (Online Appendix, Table H.4). Our empirical specifications are therefore identified on realised landings, which are post-cession: the post-cession allocation reflects the joint statutory–cession regime, not the de jure fractioning alone.

Within the industrial sector, an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) has operated since 2013: Class A licenses were allocated according to historical catch shares, whereas Class B licenses—up to 15% of the industrial allocation—have been distributed through sealed-bid first-price auctions since 2015.[2] The artisanal sector operates under a limited-entry regime, with regional TACs distributed through the *Régimen Artesanal de Extracción* (RAE) according to area, vessel size, landing site, or fisher organization.

Table 1: Statutory sectoral partition of the Central-South small pelagic TACs, pre- and post-2025 reform.

	Artisanal share of TAC (%)	
Stock	Pre-reform	2025 reform
Anchoveta (Valparaíso–Los Lagos)	78	90
Sardina común (Valparaíso–Los Lagos)	78	90
Jack mackerel (Valparaíso–Los Ríos)	10	30
Jack mackerel (Los Lagos)	10	15

Notes. Sources: SERNAPESCA (2024) for the pre-2025 regime and Congreso Nacional de Chile (2025) for the 2025 reform, in force until 31 December 2040.

Methods

Our empirical strategy targets two structural objects—climate shifters of stock productivity and fleet-specific elasticities of fishing effort—and combines them to compute counterfactual fleet responses under future climate paths. First, we identify the climate shifters: species-specific semi-elasticities of intrinsic stock productivity with respect to sea-surface temperature and chlorophyll-a anomalies. We estimate them within a Bayesian state-space model of multi-species stock dynamics, with a Schaefer law of motion and priors taken from official IFOP and SPRFMO assessments. Second, we estimate fleet-specific elasticities of annual fishing effort with respect to allocated harvest,

[2] Peña-Torres et al. (2022) document that these Class B auctions have exhibited low participation, limited capacity to reflect economies of scale, and indications of coordinated bidding.

ex-vessel prices, weather exposure, and regulatory closures, using a negative binomial count model fitted separately for the artisanal and industrial fleets, with year fixed effects that absorb aggregate non-climate shocks. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (SSP2-4.5 and SSP5-8.5, mid- and end-of-century) through two channels: the identified climate shifters apply to the stock-dynamics steady state to obtain long-run productivity (indirect biomass channel), while projected wind anomalies enter the trip equation via bad-weather days (direct weather channel).

Historical data

We use IFOP purse-seine logbook data covering 2013–2024. The dataset includes trip-level microdata with detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and vessel type, ports of departure and landing, fishery codes, haul timing and location, species composition, and retained catch. We also requested annual stock biomass series for 2000–2024 from official IFOP and SPRFMO assessments and aggregate landings by region and species from SERNAPESCA. For prices, we use ex-vessel monthly series from IFOP's *Monitoreo Económico de la Industria Pesquera y Acuícola Nacional*, which records monthly raw-material payments by processing plants to fishers, disaggregated by plant, region, species, raw-material type, and consumption class (human or animal). Processing-volume series are reported on a census basis across plants, while the price series is sample-based.

Environmental covariates are obtained from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Sea surface temperature comes from the Global Ocean Physics Reanalysis (GLORYS12V1) at 1/12° horizontal resolution (E.U. Copernicus Marine Service Information 2025c). Surface wind speed is taken from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, at 0.125° horizontal resolution and hourly frequency (E.U. Copernicus Marine Service Information 2025b). Chlorophyll-a concentrations come from the Global Ocean Colour dataset at $\sim$ 4~km horizontal resolution (E.U. Copernicus Marine Service Information 2025a). All environmental data were retrieved daily (hourly for winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 42°S (Figure 1).

The jack mackerel biomass sample over the 2000–2024 estimation window is partial: 16 of 25 years contribute a point estimate from the official IFOP hydroacoustic assessment, two more (2012, 2015) report biomass values below 3~kt—three orders of magnitude under the series' historical mean—which we treat as left-censored at 3~kt under the assumption that the acoustic survey cannot reliably distinguish small-positive values from zero, and seven carry no biomass observation. The reduced informational content is one of the structural reasons behind the non-identification result for the jack mackerel climate shifter (Section *Identification of climate shifters*).

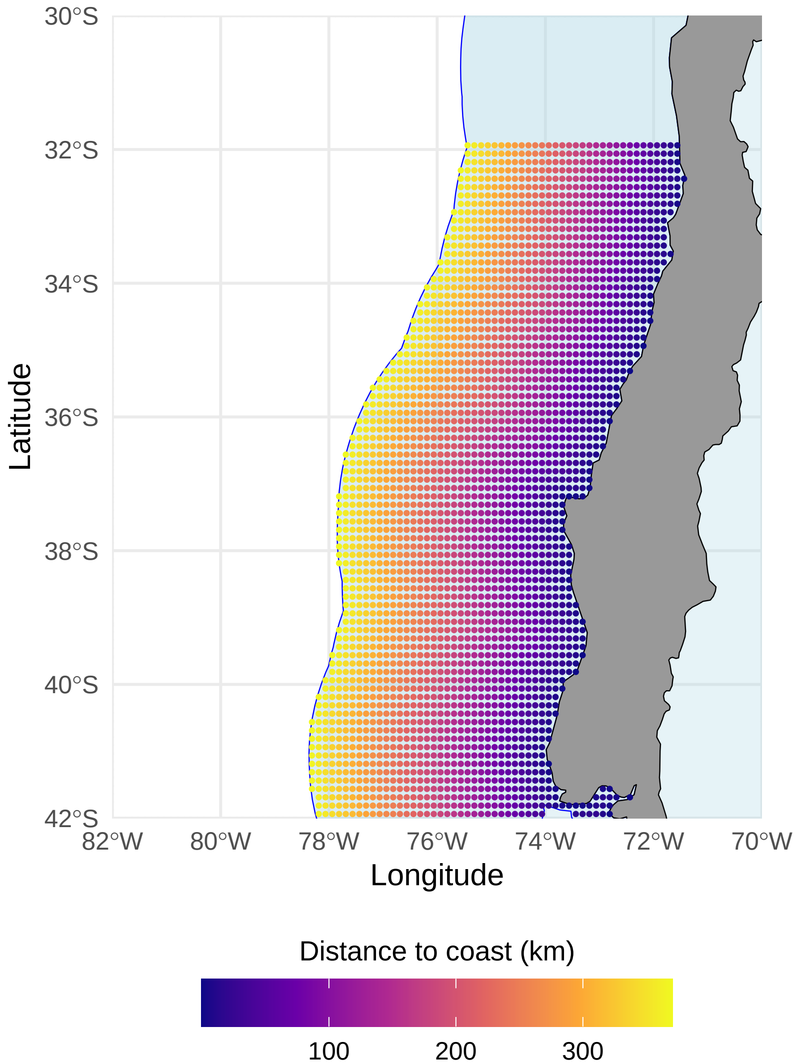

Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive using a six-model ensemble selected to span the range of warming responses across the CMIP6 archive (from low- to high-sensitivity models; full list in Online Appendix F). We use monthly outputs for two future scenarios: SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). Section *Projection approach* describes the projection horizons, baseline period, and change-factor procedure used to translate CMIP6 outputs into our empirical models.

Econometric models

Stock dynamics

Figure 2 shows that anchoveta, sardine, and jack mackerel exhibit distinct biomass trajectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once scale differences are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel in particular experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing and unfavorable environmental conditions.

Our identification target in this subsection is the vector of climate semi-elasticities $(\rho_s^{SST}, \rho_s^{CHL})$ across the three stocks $s \in \{\text{anchoveta}, \text{sardine}, \text{jack mackerel}\}$—the structural parameters that translate sea-surface temperature and chlorophyll-a anomalies into shifts in long-run stock productivity. We estimate them within a Bayesian state-space model where the latent biomass $B_{s,t}$ of each stock follows a Schaefer law of motion whose intrinsic growth rate varies log-linearly with these forcings. The law of motion is

$$B_{s,t+1} \;=\; B_{s,t} + r_{s,t}\, B_{s,t}\left(1 - \frac{B_{s,t}}{K_s}\right) \;-\; C_{s,t} \;+\; \varepsilon_{s,t}, \qquad \varepsilon_{s,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{proc},s}), \tag{1}$$

where $r_{s,t}$ is the intrinsic growth rate (specified below), K_s is the stock-specific carrying capacity, $C_{s,t}$ is the observed annual catch, and $\varepsilon_{s,t}$ is the stochastic shock to recruitment with standard deviation $\sigma_{\text{proc},s}$. The shocks are jointly distributed across species with a free 3×3 correlation

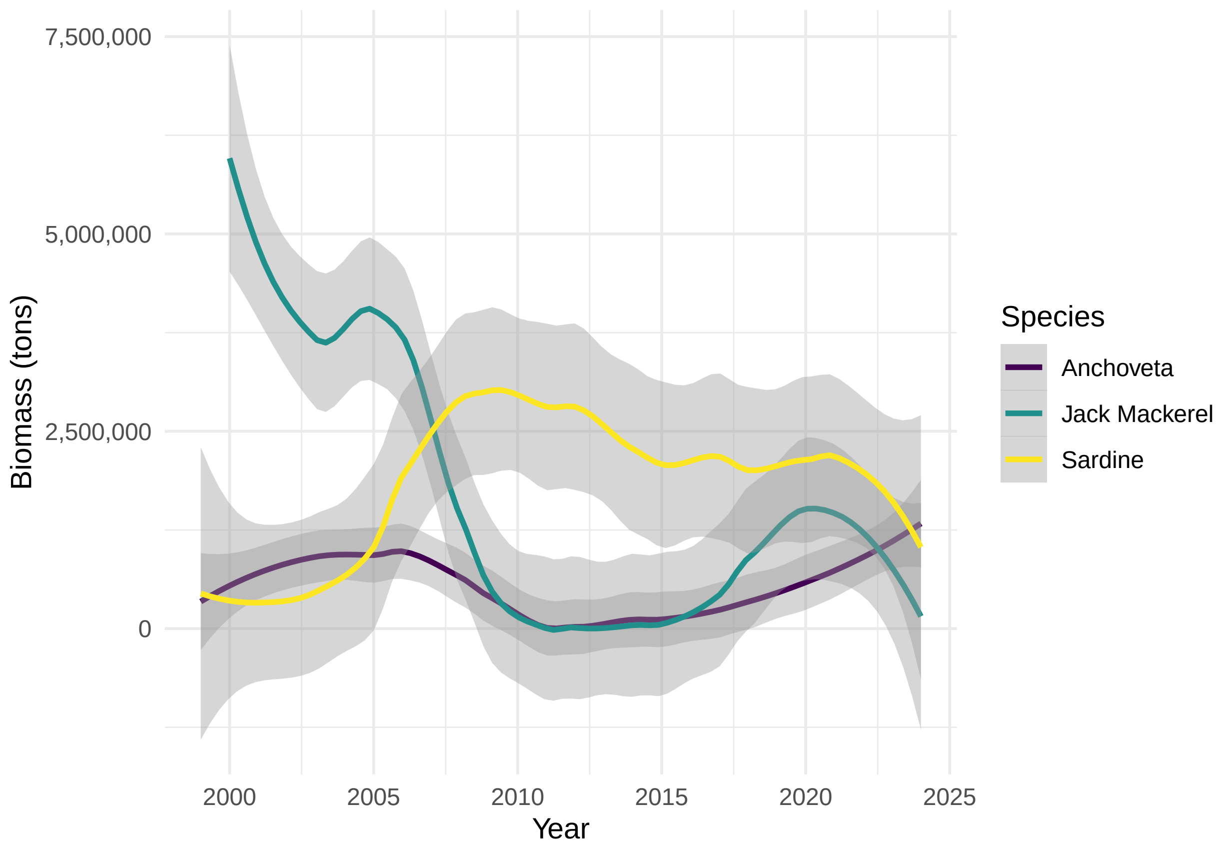

Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024). Curves are LOESS-smoothed. The jack mackerel CS series is reconstructed from the IFOP acoustic record where available, imputed from the IFOP Northern Chile series via GLM otherwise, with linear interpolation as final fallback.

matrix Ω identified from the data, allowing ecological coupling not captured by the climate forcings (e.g., direct trophic interactions or competition for plankton).

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year because environmental conditions affect recruitment and growth with a one-year delay, so that SST and CHL precede year-t biomass causally. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (Alheit and Niquen 2004; Cahuin et al. 2009; Yáñez et al. 2014).

The climate shifter on $r_{s,t}$ takes the log-linear form,

$$r_{s,t} \;=\; r_s^0 \, \exp\Big(\rho_s^{SST} \, (SST_{t-1} - \overline{SST}) \;+\; \rho_s^{CHL} \, (\log CHL_{t-1} - \overline{\log CHL}) \Big), \tag{2}$$

so ρ_s^{SST} and ρ_s^{CHL} are the semi-elasticities of intrinsic stock productivity to a one-degree Celsius SST anomaly and to a one-log-point chlorophyll-a anomaly $\log(CHL_{t-1}/\overline{CHL})$, respectively. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are generalisable—in the structural sense—to climate regimes outside the 2000–2024 estimation sample. This generalisability matters because the comparative-statics exercise of Section *Climate change projections* applies the shifter at projected SST anomalies several SDs beyond anything observed in 2000–2024.

Latent biomass is linked to the official survey-based estimates of total stock biomass $B_{s,t}^{\text{obs}}$ published by IFOP for all three species in Centro-Sur Chile, through a log-normal equation,

$$\log B_{s,t}^{\text{obs}} \;=\; \log B_{s,t} \,+\, u_{s,t}, \qquad u_{s,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{obs},s}). \tag{3}$$

where $u_{s,t}$ is the log-observation error—capturing sampling and detection noise in the hydroacoustic survey—with standard deviation $\sigma_{\text{obs},s}$.

We estimate the state-space specification by Hamiltonian Monte Carlo in Stan. With $N = 25$ annual observations per stock, joint identification of $(r_s^0, K_s, \rho_s^{SST}, \rho_s^{CHL})$ requires the prior information embedded in the IFOP and SPRFMO single-species assessments; Online Appendix A documents the prior construction and shows that maximum likelihood fails without these priors. We compare three nested specifications: (i) independent stocks with no climate shifters, (ii) joint stocks via Ω with no shifters, and (iii) joint stocks with shifters (Eq. (2)). We adopt (iii) as our primary specification; the nested predictive comparison is reported in Online Appendix B, and convergence diagnostics in Online Appendix D.

For jack mackerel we additionally fit a specification in which the growth rate is forced by the basin-scale Niño 3.4 SST anomaly at lag 1 in place of the local SST/CHL shifters, motivated by biological evidence that the Centro-Sur stock is influenced by forcing scales broader than the local coastal upwelling regime (Arcos et al. 2001; Peña-Torres et al. 2017). The posterior of that ENSO shifter is reported in Table 2 alongside the local shifters and discussed in Online Appendix E.4; anchoveta and sardine are not refit.

Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). We estimate separate effort models for the industrial and artisanal fleets to allow distinct elasticities to prices, allocated quota, weather, and biological closures. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology.

The trip equation is a negative binomial (NB) count model:

$$T_{vy} \sim \mathrm{NB}(\lambda_{vy}, \theta), \qquad \lambda_{vy} = \exp\bigl(U'_{vy}\beta\bigr)\,, \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y and θ is the dispersion parameter that accommodates the substantial overdispersion of T_{vy} (variance-to-mean ratio of 22.4 for artisanal and 4.9 for industrial).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H^{alloc}_{vy,s},\, Z_v,\, O_{vy}] \quad \text{for } s \in \{\text{anchoveta, sardine, jack mackerel}\}. \tag{5}$$

Output prices p_{sy} are species-specific ex-vessel prices constructed from IFOP's *Monitoreo Económico de la Industria Pesquera y Acuícola Nacional* by aggregating the monthly plant-level series to annual averages over peak fishing months, following Dresdner et al. (2013), and deflating to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Prices are mapped to each vessel through its modal departure region.

Quota shares do not enter Eq. (4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω^r_{vs} denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}^r_{sy}$ for species s in year y and region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H^{alloc}_{vy,s} = \omega^r_{vs}\, \bar{Q}^r_{sy}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port in the logbook records. Artisanal TACs are assigned at the regional level (Valparaíso, Biobío, Araucanía, Los Ríos, Los Lagos), while industrial TACs follow the regulatory zone structure established by SUBPESCA (the Valparaíso–Araucanía and Los Ríos–Los Lagos zones for jack mackerel; the Valparaíso–Los Lagos zone for sardine and anchoveta). This regional construction introduces cross-sectional variation in $H^{alloc}_{vy,s}$ that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

For the artisanal fleet, preliminary diagnostics revealed within-fleet heterogeneity in the responses to both allocated anchoveta and allocated sardine: vessel-type categories differ in both the sign and magnitude of these elasticities. We therefore include $H^{alloc}_{vy,\text{anchoveta}} \times$ Vessel type and $H^{alloc}_{vy,\text{sardine}} \times$ Vessel type in the artisanal specification. The estimation is restricted to Vessel type categories with at least 50 vessel-year observations (BM, LM, UNK); smaller categories (collectively $< 0.1\%$ of artisanal effort) are excluded from the fit. The jack mackerel quota does not display this within-fleet heterogeneity.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period.

First, the number of bad-weather days is computed as the count of days per year in which the maximum daily wind speed exceeds 8 m/s at the environmental grid point ($0.125° \approx 14$~km resolution) nearest to the vessel's COG.[3] Second, the number of days closed for biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in Valparaíso–Biobío; July–October in Araucanía–Los Lagos) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. The purpose of Eq. (4), however, is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation; and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock's intrinsic productivity $r_{s,t}$ through the climate shifter $(\rho_s^{SST}, \rho_s^{CHL})$ of Eq. (2) and, in turn, the law of motion in Eq. (1).

Climate projections are derived from the six-model CMIP6 ensemble described in Section *Future climate data*, under SSP2-4.5 and SSP5-8.5. We apply the change-factor method (Wilby et al. 2004; Tabor and Williams 2010) independently to each model, preserving observed interannual variability and fine-scale spatial structure from satellite records while imposing the CMIP6 climate-change signal. For each model and variable, the delta is computed between the future climatology and a historical baseline matched to the 2000–2024 estimation window. Because CMIP6 'historical' simulations terminate in 2014, we extend the baseline through 2024 using SSP2-4.5 (near-term

[3]The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. We selected this threshold via AIC comparison across 8, 10, and 12 m/s cut-offs on the trip-equation fit.

scenarios diverge negligibly).[4] We consider mid-century (2041–2060) and end-of-century (2081–2100) windows. The projections propagate two sources of uncertainty: (i) estimation uncertainty in the empirical parameters—the Bayesian posterior of the climate shifters (ρ_s^{SST}, ρ_s^{CHL}) and the maximum-likelihood standard errors of the trip-equation coefficients (β's)—and (ii) climate uncertainty from the six CMIP6 models (each gives a different projected delta). The decomposition of total variance into these two components is reported in Online Appendix F (stock productivity) and Appendix G (fleet-level trips).

For the direct weather channel, monthly wind speed deltas from CMIP6 are applied additively to the daily wind speed observations at each vessel's centre-of-gravity (COG); the number of bad-weather days per year is then recomputed at the projected mean using the same 8 m/s operability threshold as in the historical panel. Because the historical daily wind distribution at vessel COGs sits sufficiently close to the threshold, a small upward shift in the mean displaces a non-negligible mass of the upper tail past it, generating substantially more days exceeding the threshold despite the small absolute change in the mean.

For the indirect biomass channel, the CMIP6 SST delta (in °C) and log-CHL delta (in log-points) are added to the historical anomaly series. These enter Eq. (2) and shift each stock's intrinsic growth rate to $r_s^\star = r_s^0 \cdot \exp(\rho_s^{SST}\,\Delta SST + \rho_s^{CHL}\,\Delta \log CHL)$. The new $r_s^\star$ translates into a new Schaefer steady-state biomass $B_s^\star = K_s(1 - F_s/r_s^\star)$—collapsing to zero if $F_s \geq r_s^\star$. Under proportional-TAC management, each species' relative biomass shift $B_s^\star/B_s^0$ scales the corresponding vessel-level allocated harvest $H_{vy,s}^{alloc}$. The trip equation aggregates the three species-specific shifts through the Kasperski-aligned semi-elasticities β_h^s, so each vessel's projected effort response reflects all three biomass shifts simultaneously, weighted by its historical exposure to each species. We compute the climate-driven shift in long-run productivity, $\Delta r_s^\star/r_s^0$, for every combination of CMIP6 model and posterior draw of $(\rho_s^{SST}, \rho_s^{CHL})$.

Because $\Delta r_s^\star/r_s^0$ is a random variable over both dimensions (the six CMIP6 models and the posterior of the shifters), we report four summary statistics that separate the corresponding two sources of uncertainty. *Climate uncertainty*—the contribution of disagreement across CMIP6 models—is

[4] One model–variable combination is spliced from SSP5-8.5 instead because the SSP2-4.5 file is missing from the model's archive.

summarised by computing the within-model posterior median of $\Delta r_s^\star / r_s^0$ for each of the six models and then reporting the median and IQR across those six model-specific medians. *Structural uncertainty*—the contribution of parameter uncertainty about the shifters—is summarised by evaluating $\Delta r_s^\star / r_s^0$ at every posterior draw conditional on each CMIP6 model, taking the 5th and 95th percentiles to form a within-model 90% posterior credible interval, and reporting the median lower bound and median upper bound across the six models. Finally, we report the posterior probability of a negative impact, $\Pr(\Delta r_s^\star / r_s^0 < 0)$, computed within each CMIP6 model as the fraction of posterior draws yielding a negative shift and then taken as the cross-model median. Online Appendix F formalises the climate-vs-structural separation as a variance decomposition of $\Delta r_s^\star / r_s^0$ via the law of total variance; Appendix G applies the same logic to fleet-level annual trips.

Our framework is positioned as long-run comparative statics, not a year-by-year forecast. Because the empirical models are estimated on interannual variability rather than decadal trends, the projections capture short-run behavioural responses to environmental and weather shocks but cannot identify long-run adaptation by fishers or managers (Auffhammer 2018). Prices, quotas, and adaptive behaviour are held fixed at historical levels. The framework therefore provides a benchmark for the direction and magnitude of climate impacts under a fixed-institution baseline; relaxations that endogenize price adjustment, quota revision, or vessel-level adaptation are flagged as natural extensions of this work.

Results

Identification of climate shifters

Table 2 reports the posterior identification of the local climate shifters $\{\rho_s^{SST}, \rho_s^{CHL}\}_{s=1}^3$ introduced in Section *Methods*, together with the basin-scale ENSO shifter $\rho_{\text{jur}}^{\text{ENSO}}$ from the jack mackerel refit of Online Appendix E.4. The model reproduces the historical biomass trajectories within a median 90% posterior band width of approximately 20% of the mean; posterior-predictive and convergence diagnostics are reported in Online Appendices C and D. Priors are weakly informative and elicited from IFOP and SPRFMO single-species assessments (Appendix A). The posterior-to-prior standard-deviation ratio in the rightmost column measures how much the 2000–2024 biomass series refines each parameter relative to the prior; values well below one indicate identification from the likelihood.[5]

Table 2: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu, \sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\hat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardine CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jack mackerel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardine CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jack mackerel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001
ρ^{ENSO}	Jack mackerel CS	$\mathcal{N}(0.0, 0.5)$	-0.02	[-0.81, 0.80]	0.98	1.000

Note:
Shifters are semi-elasticities of intrinsic growth r_s with respect to local SST (°C), local log-CHL (log-points), and the basin-scale Niño 3.4 index. The ENSO row reports the lag-1 refit of Online Appendix E.4 for jack mackerel only; anchoveta and sardine inherit their main-specification posteriors. All $\hat{R} \leq 1.01$.

Three patterns in Table 2 organise the discussion: (i) the two coastal stocks are identified on both shifters, with sardine's SST semi-elasticity roughly twice that of anchoveta; (ii) their CHL shifters carry opposite signs; and (iii) jack mackerel returns a structural null on all three shifters, including the basin-scale ENSO refit.

First, for the two coastal upwelling stocks—anchoveta and sardine común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho^{SST}_{\text{anch}} = -1.06$ (CI $[-1.79, -0.38]$); for sardine, the corresponding semi-elasticity is more than twice as large, $\rho^{SST}_{\text{sard}} = -2.75$ (CI $[-3.46, -2.02]$). Both signs are consistent with the regional literature: the negative SST sensitivity for anchoveta aligns with the cold-water anchovy regime documented for the Humboldt Current ecosystem (Alheit and Niquen 2004; Schwartzlose et al. 1999), and the stronger sardine sensitivity matches the negative coupling between SST anomalies and *Strangomera bentincki* recruitment in Centro-Sur Chile reported by Gomez et al. (2012) with $r = -0.83$. Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26\,^{\circ}$C over 2000–2024), a thermal

[5] The ratio captures the relative information content of the data, isolating identification from prior centring: a precise likelihood shrinks the posterior dispersion around wherever the data locate the parameter, whereas a non-informative likelihood leaves the prior dispersion unchanged regardless of how the posterior mean moves.

shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardine by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of $+1\,^{\circ}$C, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardine. Applied to a fishery whose anchoveta–sardine combined Centro-Sur TAC has averaged roughly 440 kt per year over the 2012–2024 window, the implied losses are on the order of 300–400 kt per year under sustained $+1\,^{\circ}$C warming. The caveat is that $+1\,^{\circ}$C lies approximately four historical standard deviations beyond the 2000–2024 estimation window, so the projection relies on the log-linear extrapolation embedded in the shifter function.

Second, the two coastal stocks respond to chlorophyll-a with opposite signs: $\rho^{CHL}_{\text{anch}} = -3.64$ (CI $[-5.01, -2.25]$) versus $\rho^{CHL}_{\text{sard}} = +2.17$ (CI $[+0.96, +3.37]$). At one historical standard deviation of $\log CHL$ ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock lowers anchoveta growth by approximately 29% while raising sardine growth by approximately 23%. The opposite-signed CHL semi-elasticities in Table 2 are consistent with the well-documented anchoveta–sardine asymmetry under regime shifts in the Humboldt Current ecosystem (Alheit and Niquen 2004; Schwartzlose et al. 1999): they might reflect the relative competitive dynamics between the two species under high-productivity regimes rather than a direct biological effect of chlorophyll on anchoveta. The positive coupling between coastal chlorophyll and common sardine recruitment in Centro-Sur Chile is documented with correlation $r = 0.92$ between CHL and log recruitment by Gomez et al. (2012). A warming-and-greening regime therefore does not produce a uniform shock on the artisanal fleet, because anchoveta and sardine respond to chlorophyll with opposite signs. The net impact on each vessel depends on its realized catch composition over the two species, which we integrate at the fleet level in Section *Climate change projections*.

Third, and in marked contrast, neither SST nor CHL shifters are identified for jack mackerel. Unlike anchoveta and sardine, jack mackerel priors are vague and centered at zero (Appendix A), so the posterior moves only if the data refines it. That the posterior coincides with the prior—$\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$, means near zero, credible intervals straddling zero—indicates that the 2000–2024 biomass series provides insufficient information to identify climate effects on jack mackerel productivity at this scale, consistent with jack mackerel ranging across the broader Southeast

Pacific rather than being driven by local Chilean conditions (it is managed through SPRFMO accordingly). The model correctly reports this non-identification rather than producing a spurious coefficient.

Online Appendix E reports five complementary robustness tests of this null: three nested coastal aggregations, a dual-source extension with a Northern Chilean acoustic series, basin-scale ENSO shifters at lags 1 and 2, a joint specification with all three shifters active simultaneously, and an extension using the broader SPRFMO range-wide jack mackerel stock assessment. All five converge on posterior-to-prior ratios between 0.94 and 1.03, indistinguishable from unity. These tests try to rule out the standard technical explanations for the null: too small an area, too few years, the wrong climate variable, the wrong econometric specification, or the wrong geographic scope. The relevant climate forcing for the Centro-Sur jack mackerel stock operates at margins—behaviour, location, and the seasonal timing of spawning and migration (Arancibia et al. 2019; Arcos et al. 2001; Peña-Torres et al. 2017)—not captured in the elasticity of annual aggregate productivity.

Figure 3 visualises the posterior–prior updating as a forest plot.

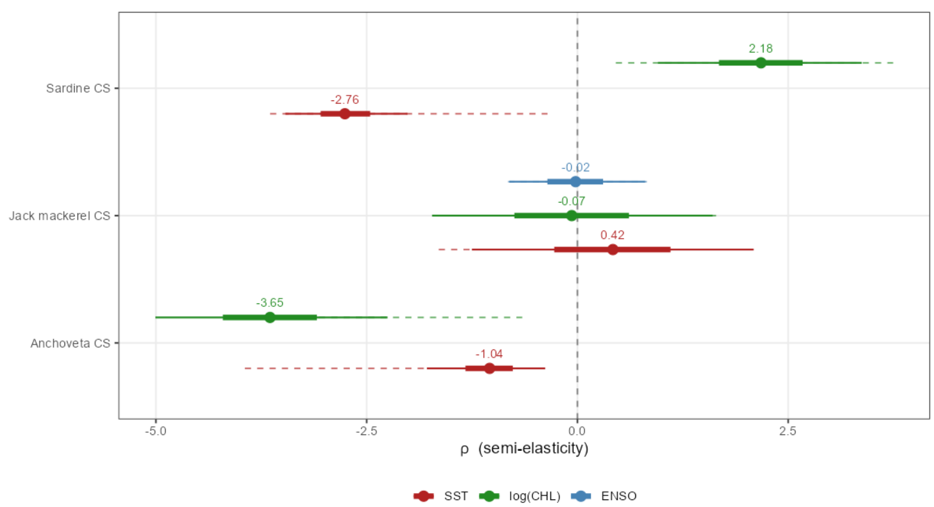

Figure 3: Posterior and prior distributions of the climate shifters ρ_s^{SST}, ρ_s^{CHL}, and ρ_{jur}^{ENSO} by stock. Solid lines show the posterior 90% (thin) and 50% (thick) credible intervals; dashed lines show the prior 90% credible interval. Anchoveta and sardine show substantial posterior updating with narrow credible intervals excluding zero. Jack mackerel posteriors (SST, CHL, ENSO) coincide with their vague priors centred at zero, indicating non-identification.

Total annual trips

Table 3 reports the negative binomial estimates separately for each fleet segment. The specification includes year fixed effects, which absorb aggregate shocks affecting both fleets contemporaneously—most notably the 2019 social outbreak (*estallido social*) and the 2020–2022 COVID-19 period—and identify the trip-equation coefficients from within-year cross-vessel variation.

Table 3: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4184$)

	Dependent variable:	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m^3)	0.097 (0.221)	0.132*** (0.045)
Allocated harvest anchoveta ($H^{alloc}_{vy,anch}$, tons)	−0.0003 (0.0002)	0.007*** (0.002)
Allocated harvest sardine ($H^{alloc}_{vy,sard}$, tons)	0.0001*** (0.00003)	0.004*** (0.001)
Allocated harvest jack mackerel ($H^{alloc}_{vy,jur}$, tons)	0.00002* (0.00001)	0.001 (0.0005)
Price anchoveta (1000s $/ton)	−0.004 (0.004)	0.010*** (0.003)
Price sardine (1000s $/ton)	−0.016*** (0.003)	0.004** (0.002)
Price jack mackerel (1000s $/ton)	0.008*** (0.003)	0.007*** (0.002)
Bad weather days	−0.0003 (0.001)	−0.001 (0.0005)
Veda days	−0.079*** (0.005)	0.012*** (0.002)
Vessel type (UNK)	−1.450*** (0.163)	0.166 (0.102)
Vessel type (LM)		0.466*** (0.100)
$H^{alloc}_{vy,anch}\times$ Vessel type (LM)		−0.007*** (0.002)
$H^{alloc}_{vy,anch}\times$ Vessel type (UNK)		−0.007*** (0.002)
$H^{alloc}_{vy,sard}\times$ Vessel type (LM)		−0.003*** (0.001)
$H^{alloc}_{vy,sard}\times$ Vessel type (UNK)		−0.00002 (0.001)
Constant	15.763*** (2.605)	−3.776*** (0.519)
Year fixed effects	Yes	Yes
Pseudo-R^2 (Cragg-Uhler)	0.302	0.632
Observations	319	4,184

Note: *p<0.1; **p<0.05; ***p<0.01. Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton.
LR test NB vs Poisson: Industrial $\chi^2 = 397.6$; Artisanal $\chi^2 = 36496.9$ (both $p < 0.001$).

The coefficients on species-specific allocated harvest reveal cross-fleet heterogeneity. For the industrial fleet, the sardine allocation yields a positive and significant trip response ($p < 0.001$), the jack mackerel allocation is positive and marginally significant ($p < 0.10$), and the anchoveta coefficient is not statistically distinguishable from zero. The artisanal fleet shows positive and significant responses to anchoveta and sardine allocations (both at $p < 0.001$); the jack mackerel allocation is positive but not statistically distinguishable from zero. Both species show scale-dependent responses in the artisanal fleet, but with different patterns. For anchoveta, only the *bote motor* (BM, 8% of the panel) exhibits a positive marginal coefficient ($\hat{\beta}_h^{\text{anch,BM}} = +0.007$, $p < 0.01$); the *lancha motor* (LM, 77%) and *unidentified* (UNK, 14%) categories cancel this base effect through significant negative interactions (-0.007 each, both $p < 0.01$). For sardine, BM and UNK exhibit positive coefficients ($\hat{\beta}_h^{\text{sard,BM}} = +0.004$, $p < 0.01$; UNK interaction ≈ 0); only LM cancels the base effect through a significant negative interaction (-0.003, $p < 0.01$).

The two fleets differ in price responsiveness. For the industrial fleet, both the jack mackerel and sardine prices are significant at the 1% level—jack mackerel with a positive coefficient and sardine with a negative coefficient; the anchoveta price has no detectable effect. The artisanal fleet shows positive and significant coefficients on all three prices: anchoveta and jack mackerel at the 1% level and sardine at the 5% level. The artisanal pattern is consistent with a standard supply-side response: higher ex-vessel prices for the species the fleet targets are associated with more trips. The negative industrial sardine-price coefficient is contrary to a supply response, but the trip equation is not a causal demand specification—prices, harvest allocations, and biomass are jointly determined within the panel—and prices enter only as control variables that absorb year-on-year market variation.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but it is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns. Adverse weather days carry a negative coefficient for both fleets but are not statistically distinguishable from zero. We retain these point estimates for the direct weather channel of Section *Projection approach*; their estimation uncertainty propagates into the cross-model spread of Table 6. Biological closure days are strongly negative for the industrial fleet. The artisanal coefficient is positive but admits no causal reading under year fixed effects: the underlying closure variable is constant within each artisanal management zone, so the regression identifies only off the cross-zone differential and conflates locational heterogeneity in artisanal effort intensity with the causal effect of closures.

Climate change projections

The projection methodology is summarized in Section *Projection approach*. Under each scenario and time window, we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days that enter the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r^{\star}_{s,t}$, obtained by evaluating the identified climate shifter of Eq. (2) at the projected SST and log-CHL anomalies (indirect channel).

Projected environmental changes

Table 4 summarises the projected environmental changes as cross-model medians with cross-model interquartile range (IQR). SST warms by +0.6 to +2.4 °C across scenarios. The chlorophyll-a cross-model median is close to zero in all scenario-window pairs (between −1.3% and +2.2%), while the cross-model IQR is substantially wider (up to −7% to +3% under SSP5-8.5 end-of-century), reflecting documented CMIP6 disagreement on primary-productivity responses. Wind speed increases modestly (+0.1 to +0.4 m/s cross-model median), but translates into +8.5 to +23.5 additional bad-weather days per year because the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that a small upward shift in the mean displaces a non-negligible mass of the upper tail past it.

Table 4: Projected environmental changes for Centro-Sur Chile (CMIP6 ensemble, change-factor method; cross-model median with cross-model IQR in brackets).

SSP	Window	n_m	Δ SST (°C)	Δ CHL (%)	Δ wind speed (m/s)
SSP2-4.5	2081–2100	5	+1.31 [+0.84, +1.56]	+0.0 [−3.6, +1.4]	+0.19 [+0.10, +0.24]
SSP2-4.5	2041–2060	5	+0.58 [+0.50, +0.93]	+2.2 [+0.0, +2.8]	+0.14 [+0.13, +0.15]
SSP5-8.5	2081–2100	6	+2.39 [+1.93, +3.29]	−1.3 [−7.4, +3.0]	+0.42 [+0.36, +0.45]
SSP5-8.5	2041–2060	6	+0.92 [+0.61, +1.23]	+0.7 [−2.3, +3.4]	+0.21 [+0.14, +0.32]

Note:
SSP2-4.5 panels use $n_m = 5$ because one of the six CMIP6 models in the ensemble does not publish chlorophyll-a projections for SSP2-4.5; this matches the sub-ensemble used in Table 5.

Comparative statics on intrinsic stock productivity

Table 5 reports the four summary statistics of $\Delta r^{\star}_s / r^0_s$ defined in Section *Projection approach*, for each stock under each scenario-window pair. The variance decomposition in Online Appendix F shows that the two sources contribute asymmetrically across stocks: climate uncertainty dominates for sardine (85–91% of total variance) while structural uncertainty dominates for anchoveta (56–88% within-model). Jack mackerel is reported as non-identified ("n.i.' ') consistent with the identification result in Section *Identification of climate shifters*.

Under the moderate SSP2-4.5 mid-century scenario (cross-model median $+0.58\,^{\circ}$C; $\Delta \log \mathrm{CHL}$ near zero in the cross-model median, with non-trivial cross-model spread), the cross-model median of anchoveta's intrinsic productivity declines by 51% and sardine's by 79%, with posterior probability of negative impact equal to one in both cases.

Under the high-emissions SSP5-8.5 end-of-century scenario (cross-model median $+2.39\,^{\circ}$C, $\Delta \log \mathrm{CHL}$ cross-model median -0.013 with 5th-percentile of -0.30 across models) the cross-model medians deepen to -90% and -99.9% respectively, and the entire 90% credible interval of sardine's response lies below zero in every model of the ensemble. The anchoveta cross-model probability of decline is 0.99 rather than 1 because under a small fraction of (model, draw) pairs a sufficiently negative $\Delta \log \mathrm{CHL}$ combined with $\rho^{CHL}_{\mathrm{anch}} < 0$ offsets the SST-driven collapse.

The identified signs confirm the qualitative reading of the shifters in Section *Identification of climate shifters*: both coastal stocks respond negatively to SST, but sardine carries roughly twice the SST semi-elasticity of anchoveta, and its CHL semi-elasticity is of the opposite sign, so that the compound response—negative thermal shock combined with negative primary-productivity shock—is sharper for sardine than for anchoveta.

The comparative-statics exercise is a long-run object rather than a year-by-year forecast. It describes the steady-state intrinsic productivity of each stock under a counterfactual climate, *holding fixed* the law of motion and the fishery's regulatory regime. The shifters $(\rho_s^{SST}, \rho_s^{CHL})$ enter the law of motion as coefficients on exogenous climate forcings, which makes them invariant to the joint distribution of (SST, B, C) and therefore evaluable at climate regimes outside the estimation window—in particular, at an SST anomaly of $+2.3$°C that lies roughly nine historical standard deviations beyond the 2000–2024 mean. A purely correlational specification would not license this extrapolation.

Table 5: Comparative statics on intrinsic stock productivity under the CMIP6 six-model ensemble. SSP2-4.5 panels use $n_m = 5$ because one of the six models does not publish chlorophyll-a projections for that scenario; SSP5-8.5 panels use all six.

					Summaries of $\Delta r_s^\star / r_s^0$			
Stock	Scenario	n_m	ΔSST (°C)	$\Delta \log \text{CHL}$	Cross median	Cross-model IQR	Within posterior 90% CI	$\Pr(\Delta < 0)$
Anchoveta CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-50.7%	[-52.7%, -47.8%]	[-68.0%, -27.0%]	1.00
Anchoveta CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-71.0%	[-76.3%, -60.4%]	[-91.1%, -31.0%]	1.00
Anchoveta CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-62.4%	[-68.0%, -55.9%]	[-80.9%, -31.2%]	1.00
Anchoveta CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-89.6%	[-91.5%, -87.4%]	[-98.3%, -54.2%]	0.99
Sardine CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-78.6%	[-93.2%, -72.5%]	[-85.7%, -67.3%]	1.00
Sardine CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-97.2%	[-98.9%, -89.8%]	[-98.9%, -92.6%]	1.00
Sardine CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-91.1%	[-96.7%, -78.5%]	[-95.1%, -83.2%]	1.00
Sardine CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-99.9%	[-100.0%, -99.4%]	[-100.0%, -99.3%]	1.00
Jack mackerel CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	n.i.	n.i.	n.i.	n.i.

Note:
Entries summarise the posterior of the proportional change in intrinsic productivity, integrated over (rho_SST, rho_CHL) from the full state-space specification. Cross-model IQR is the 25th-to-75th percentile range across CMIP6 models of within-model posterior medians (climate uncertainty). Within posterior 90% CI is the cross-model median of within-model 5th-to-95th-percentile posterior intervals (structural uncertainty, climate fixed). Jack mackerel CS is reported as non-identified ('n.i.'); see Results identification section. Raw numerical values and model-level disaggregation are in tables/growth_comparative_statics_raw.csv and ..._by_model.csv; Appendix F reports the variance decomposition.

Implications for fleet-level effort

Table 6 translates the comparative-statics object on intrinsic stock productivity in Table 5 into a comparative-statics object on fleet-level annual trips. Each posterior draw is propagated through the stock-dynamics steady-state equation under historical average fishing pressure and then through the negative binomial trip equation of Section *Total annual trips*. Vessel-level responses are then aggregated within fleet, with each vessel weighted by its 2012–2024 catch composition.

Table 6: Long-run implications of CMIP6 climate scenarios for fleet-level fishing effort, under a Schaefer steady-state thought experiment with fishing pressure fixed at the 2000–2024 historical average. Entries are cross-model medians with cross-model IQRs in brackets, aggregated within each CMIP6 model over posterior draws and vessels within fleet.

Fleet	Scenario	$\Pr(f_v^H < 0.5)$	%Δ trips (marginal)	%Δ trips (conditional)
Artisanal	SSP2-4.5, 2041-2060	0.95 [0.93, 0.96]	-16.1% [-16.9%, -14.9%]	-8.0% [-8.2%, -7.7%]
Artisanal	SSP2-4.5, 2081-2100	0.98 [0.97, 0.99]	-17.0% [-17.1%, -16.9%]	-7.9% [-13.8%, -5.2%]
Artisanal	SSP5-8.5, 2041-2060	0.98 [0.96, 0.98]	-16.6% [-16.8%, -15.9%]	-8.8% [-11.2%, -6.5%]
Artisanal	SSP5-8.5, 2081-2100	0.99 [0.99, 0.99]	-17.6% [-17.8%, -17.5%]	-6.2% [-10.6%, -5.9%]
Industrial	SSP2-4.5, 2041-2060	0.12 [0.11, 0.12]	-1.0% [-1.0%, -0.8%]	-0.7% [-0.7%, -0.7%]
Industrial	SSP2-4.5, 2081-2100	0.12 [0.12, 0.12]	-0.9% [-1.0%, -0.8%]	-0.7% [-0.8%, -0.7%]
Industrial	SSP5-8.5, 2041-2060	0.12 [0.12, 0.12]	-0.8% [-0.9%, -0.8%]	-0.7% [-0.9%, -0.7%]
Industrial	SSP5-8.5, 2081-2100	0.12 [0.12, 0.12]	-1.0% [-1.1%, -1.0%]	-1.0% [-1.0%, -0.9%]

Note:
Entries summarise the posterior of the proportional change in fleet-level annual trips, computed for each of the six CMIP6 models in the ensemble and then aggregated across models. Within each model, the posterior is integrated over draws of the full state-space specification and vessels within fleet (composition-weighted exposure). Cross-model medians and 25th-to-75th-percentile inter-quartile ranges across the n_m models are reported in brackets; the median across models of the within-model 5th-to-95th-percentile posterior interval (within posterior CI) is archived in tables/trip_comparative_statics_raw.csv. f_H is the composition-weighted biomass factor relative to the historical baseline: f_H = sum over species of omega_vs * B_star_s / B_hist_s, where B_star_s = K_s * (1 - F_hist_s / r_eff_s) is the Schaefer steady-state biomass and omega_vs is the vessel's realized catch share in species s. Pr(f_H < 0.5) reports the posterior probability that the vessel's composition-weighted biomass factor falls below one-half. The marginal median integrates over all posterior draws within each model; the conditional median restricts to draws with f_H >= 0.5, isolating the pure climate-elasticity component from the steady-state collapse component of the trip response. Jack mackerel is treated as non-identified (factor_B = 1) consistent with the identification discussion in the Results section. Raw numerical values, by-model disaggregation, and by-stock extinction probabilities are archived in tables/trip_comparative_statics_raw.csv, tables/trip_comparative_statics_by_model.csv, and tables/trip_comparative_statics_extinct.csv. The narrow cross-IQR observed in many cells reflects a floor effect: once Schaefer steady-state collapse saturates the portfolio-weighted biomass factor across models, the trip response converges to exp(-beta * H_alloc * 1), independent of climate magnitude. Appendix G reports the formal two-way variance decomposition of the fleet-level trip response, which confirms that the within-model component (posterior plus vessel heterogeneity) dominates the total variance under saturated regimes; Appendix F reports the analogous decomposition for the underlying productivity response.

The posterior probability of portfolio loss—defined as a reduction of more than fifty percent in the composition-weighted biomass factor f_v^H relative to the 2000–2024 historical baseline—exceeds

0.95 for the artisanal fleet under every CMIP6 scenario considered and reaches 0.99 under SSP5-8.5 end-of-century. For the industrial fleet, the loss probability is a stable 0.12 across all four scenarios, driven entirely by its five-percent exposure to sardine; the remaining 95% of its historical portfolio is allocated to jack mackerel, whose climate shifter is treated as non-identified in Section *Identification of climate shifters*. The asymmetry in loss probability between the two fleets is first-order and stable across climate pathways, approximately eight-fold across all four scenarios.

Decomposing the fleet-level trip response isolates the channels through which this asymmetry operates. The *marginal* posterior median of %Δ trips ranges from -16.1% (SSP2-4.5 mid) to -17.6% (SSP5-8.5 end) for the artisanal fleet and from -0.8% to -1.0% for the industrial fleet—a ratio of roughly seventeen to one at end-of-century. The *conditional* posterior median, restricted to draws with $f_v^H \geq 0.5$, isolates the climate-elasticity component from the portfolio-collapse component and yields -6.2% to -8.8% for artisanal and -0.7% to -1.0% for industrial, a conditional ratio of roughly seven to nine.

Two channels generate this asymmetry. The *indirect biomass channel* transmits the climate signal asymmetrically through portfolio composition: the artisanal fleet is concentrated in the two coastal-upwelling stocks with identified climate shifters, while 95% of the industrial fleet's portfolio sits in jack mackerel, whose Centro-Sur shifter is non-identified and held at the climatological mean.

The *direct weather channel* transmits the signal through differential weather sensitivity: $\widehat{\beta}^{\text{ART}}_{\text{weather}} = -0.0007$ vs $\widehat{\beta}^{\text{IND}}_{\text{weather}} = -0.0003$ (both indistinguishable from zero under vessel-clustered SEs), translating the projected $+10$ to $+23.5$ days of additional bad weather into a -0.7% to -1.5% contribution to the artisanal response and near-zero for industrial. The limited cross-sector transferability of the prevailing sector-level allocation locks this dual exposure into differential losses of harvest capacity, consistent with the portfolio mechanism documented by Kasperski and Holland (2013) and the access constraints of Oken et al. (2021).

The between-fleet asymmetry decomposes further into within-artisanal heterogeneity by vessel type (Table 7). The dominant *lancha motor* (LM) segment (77% of the panel) contracts by -19.6% under SSP5-8.5 end-of-century. Although LM has near-zero marginal coefficients on both anchoveta and sardine after interaction cancellation, its absolute exposure to the collapsing sardine stock dominates the contraction ($H^{\text{alloc}}_{\text{sard}}$ median $\approx$ 616 t/yr, an order of magnitude above the other categories). The smaller *bote motor* (BM) segment (8%) contracts by −10.0% at the marginal median, less than LM because its sardine exposure (≈ 66 t/yr) is roughly ten times smaller. The intermediate UNK segment (14%) contracts by −11.9%. Portfolio-loss probability exceeds 0.98 in all three categories, so the differential within artisanal is concentrated in the magnitude rather than the incidence of portfolio collapse.

Table 7: Within-artisanal-fleet decomposition of the long-run trip response by vessel type, under SSP5-8.5 end-of-century. Entries are cross-model medians with cross-model IQRs in brackets. The marginal column integrates over all posterior draws; the conditional column restricts to draws in which the composition-weighted biomass factor is at least one-half, isolating the climate-elasticity component from the steady-state collapse component.

	Segment	Pr(portfolio loss)	%Δ trips (marginal)	Cross-model IQR	%Δ trips (conditional)
LM	Lancha motor (LM)	0.99	-19.6%	[-19.7%, -19.5%]	-16.0%
UNK	Unidentified (UNK)	0.98	-11.9%	[-12.0%, -11.8%]	-2.1%
BM	Bote motor (BM)	1.00	-10.0%	[-10.2%, -9.8%]	-5.6%

Note:
Categories with fewer than 50 vessel-year observations (BR, BRV, L) collectively account for less than 0.1 percent of artisanal effort and quota and are excluded from this table. Vessel type BM is the reference category in the negative binomial interaction; LM and UNK enter as deviations. Raw cross-model values and other scenario-window combinations are archived in trip_comparative_statics_by_tipo_emb_raw.csv.

Three caveats qualify this thought experiment. First, the Schaefer steady-state comparative statics holds fishing pressure fixed at the 2000–2024 historical average; any regulatory response that lowered F in the face of falling productivity would attenuate both the loss probability and the marginal trip response reported here. Second, the 2000–2024 window was not itself in Schaefer equilibrium under F_{hist}: an internal consistency check evaluated at $\Delta SST = 0$ and $\Delta \log CHL = 0$ yields a median f_v^H of 1.015 rather than 1.000, so the results should be read as responses *relative to the baseline steady-state*, not relative to the observed 2000–2024 trajectory. Third, jack mackerel is treated as non-identified at the Centro-Sur scale, a null on *local identification* rather than on biological climate sensitivity: the 25-year window provides only 16 informative annual observations against a historical SST envelope of $\hat{\sigma}_{SST} \approx 0.26\,^{\circ}\mathrm{C}$. Five complementary tests reported in Online Appendix E (nested coastal domains, dual-source extension, basin-scale ENSO, joint specification, SPRFMO integration) all return posterior-to-prior ratios indistinguishable from unity, indicating that the local null is structural. The industrial fleet's apparent climate protection in Table 6 therefore rests on a null about local identification, which a range-wide SPRFMO analysis would be the appropriate setting to refine.

Discussion

Our results reveal a substantial asymmetry in how climate change affects different segments of Chile's small pelagic fishery, governed by the differential exposure of each fleet's species portfolio to the identified climate parameters under limited cross-sector transferability. The artisanal fleet, concentrated in the sardine–anchoveta pair, is the segment whose long-run harvest capacity is most exposed to the CMIP6 signal; the industrial fleet's 95% exposure to jack mackerel (whose local productivity is non-identified) buffers it under the maintained convention. This finding aligns with the broader literature on heterogeneous climate impacts within fleets (Sumaila et al. 2011; Free et al. 2019) and underscores that aggregate projections can mask first-order distributional consequences inside the same fishery. The asymmetry further decomposes within the artisanal fleet by vessel type: the dominant LM segment contracts by approximately twenty percent, while the smaller BM segment contracts by approximately ten percent (Table 7).

The non-identification of $(\rho_{\text{jur}}^{SST}, \rho_{\text{jur}}^{CHL})$ at the Centro-Sur scale is itself a substantive result rather than a nuisance, but its policy reading depends on the distinction between *non-identification* and *no climate effect*. Five converging tests reported in Online Appendix E (varying coastal aggregation, augmenting biomass coverage, replacing local shifters with basin-scale ENSO, joint specification, and SPRFMO integration) all return posterior-to-prior ratios in 0.94–1.03. Existing empirical evidence documents that climate forcing operates on jack mackerel at scales broader than local coastal anomalies-\/-\/-@Arcos2001-jq on intrusion of the 15 °C isotherm during the 1997–98 El Niño, Peña-Torres et al. (2017) on ENSO-driven shifts in location choices in the Chilean fishery—but each documents climatic effects on *margins* (location choice, range distribution, spawning timing) that are not necessarily captured in the elasticity of annual aggregate productivity. Closing the identification gap will require either a longer biomass record or a basin-scale SPRFMO assessment with explicit cross-stock spillover. We accordingly hold $r^*_{\text{jack mackerel}}$ fixed in the projections; a prior-propagation sensitivity reported in Online Appendix E.4 returns a 90% predictive interval of approximately three orders of magnitude on the productivity factor under SSP5-8.5 end-of-century, confirming that the fixed-r^* convention is the defensible reporting choice.

The two channels through which climate change reaches fleet-level effort operate on very different scales and on different segments of the fleet. The direct weather channel is small in mean wind anomaly—the cross-model median CMIP6 increase in near-surface wind speed is below +0.5 m/s even under SSP5-8.5 end-of-century—but the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that additional bad-weather days exceed twenty per year on average under SSP5-8.5 end-of-century. This translates into a −0.7% to −1.5% marginal contribution to the artisanal trip response and a contribution indistinguishable from zero for the industrial fleet, reflecting the differential weather sensitivity captured by $\hat{\beta}_{\text{weather}}$. The indirect biomass channel dominates the response in absolute terms for both fleets but operates entirely through portfolio composition, so the asymmetry it generates is intrinsic to the fixed sector-level allocation of stocks across segments. Climate adaptation policy in the Centro-Sur therefore needs to address both margins. The indirect channel calls for reform of the institutional architecture that mediates portfolio exposure—quota allocation rules, cross-fleet transferability, and the design of TACs under shifting species productivity. The direct channel calls for operational measures targeted at the artisanal segment specifically, such as expanded port infrastructure, vessel reinforcement programs, or weather-indexed insurance, which are first-order for that fleet but redundant for the industrial purse-seine fleet. This conclusion is based on the CMIP6 monthly wind ensemble; downscaled regional climate models that resolve extreme wind events could only raise the weight of the direct channel further.

CMIP6 SST anomalies under SSP5-8.5 end-of-century reach roughly +2.3 °C, nearly an order of magnitude outside the 25-year estimation window ($\hat{\sigma}_{SST} \approx 0.26\,^{\circ}$C). The corresponding posterior of $\Delta r_s^{\star}/r_s^0$ therefore leans on the log-linear extrapolation embedded in the shifter function. Thermal-tolerance arguments in the climate-fisheries literature (Cheung et al. 2010; Free et al. 2019) imply that for stocks operating below their thermal optimum, a log-linear specification, if anything, *understates* the productivity response under high warming, so the projected magnitudes should be read as conservative within the class of monotone shifter functions consistent with the identified posterior signs.

Seven caveats qualify the present analysis: holding ex-vessel prices and management rules constant; substantial cross-model spread in chlorophyll-a; identification on a 25-year series limiting non-linearity detection; the absence of risk preferences in the trip equation; the zonal coding of biological closures; the purse-seine focus of the trip-equation panel; and the missing wind components for CESM2 in the direct weather channel. Online Appendix I states each in full and qualifies its implication for the interpretation developed above.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota-allocation policy. A natural first reading would attribute the asymmetry to rigid single-species permits in the artisanal sector, but both sectors operate *de jure* multi-species permit portfolios: SUBPESCA's 2012 registry records simultaneous authorisations across all three pelagic species for 76% of industrial vessels (Subsecretaría de Pesca y Acuicultura (SUBPESCA) 2026); the IFOP logbook in turn reveals that 90% of artisanal vessels operate a multi-species portfolio in any given year. The asymmetric climate sensitivity we identify is therefore not driven by permit count but by four dimensions of the institutional architecture. First, the statutory partition fixes inter-sectoral shares at the legislative level (Table 1). Second, the partition admits a single administratively rationed re-allocation channel: industrial titulares cede 85–99% of their assigned anchoveta and sardine TACs to artisanal cessionaries each year (25–77 kt per species-year), while jack mackerel cessions are an order of magnitude smaller and reverse direction across years (Online Appendix H.3). The channel is non-market in the conventional sense but operational rather than dormant. Third, within-sector transferability is asymmetric: industrial coefficients trade under LTP since 2013, while artisanal RAE assignments remain locked to the regional registry. Fourth, the revealed portfolio composition differs in its climate correlation structure: the artisanal logbook shows 80–86% of activity concentrated in the anchoveta–sardina pair (both with identified negative climate elasticities), while 67% of industrial activity sits in the same pair with the remainder in jack mackerel and caballa (non-identified at the Centro-Sur scale). The asymmetry the paper estimates therefore quantifies the cost of a statutory partition combined with a climate-correlated artisanal portfolio under projected SSP5-8.5 stress. Online Appendix H reports the descriptive evidence and Online Appendix H.2 computes a policy counterfactual.

Implications of the 2025 fractioning reform

The asymmetry estimated in this paper is identified on the pre-reform fractioning regime that prevailed during the estimation window (2013–2024). The 2025 reform, in force from 2026 until

2040, modifies that regime in two relevant directions for the Central-South small pelagic fishery (Congreso Nacional de Chile 2025). First, the artisanal share of the anchoveta and sardine TACs rises from 78% to 90% across the Valparaíso–Los Lagos macrozone; the same artisanal sector therefore now holds a deeper claim on stocks whose long-run productivity is projected to collapse under SSP5-8.5 end-of-century. Second, the artisanal share of the jack mackerel TAC rises from 10% to 30% in Valparaíso–Los Ríos and from 10% to 15% in Los Lagos—a quota-weighted aggregate of approximately 28% across the macrozone—giving the artisanal sector partial access to the species whose local productivity is non-identified at the Centro-Sur scale and whose climate response is held fixed in our projections. Table 8 reports the static plug-in counterfactual at SSP5-8.5 end-of-century, holding the structural posterior and the TAC level at 2024 values and varying only the statutory artisanal share.

Table 8: Static counterfactual: artisanal absolute landing under the pre-reform and 2025 fractioning regimes at SSP5-8.5 end-of-century, Centro-Sur (Valparaíso–Los Lagos), 2024 Cuotas Globales.

		Artisanal share (α_s)		Artisanal landing L_{art} (kt	
Stock	TAC 2024 (kt)	Pre-reform	2025 reform	Pre-reform	2025 reforn
Anchoveta (Valparaíso–Los Lagos)	211.3	0.78	0.90	17.1	19.
Sardina común (Valparaíso–Los Lagos)	290.6	0.78	0.90	0.2	0.
Jack mackerel (Valparaíso–Los Ríos)	578.6	0.10	0.30	57.9	173.
Jack mackerel (Los Lagos)	80.6	0.10	0.15	8.1	12.
Total Centro-Sur				**83.3**	**205.**

Notes. Static plug-in evaluation of $L^{(R)}_{\text{art}} = \sum_s \alpha^{(R)}_s \cdot (1 + \widehat{\Delta r^\star_s / r^0_s}) \cdot Q^0_s$, with $\widehat{\Delta r^\star_s / r^0_s}$ set to the cross-model median at SSP5-8.5 end-century from Table 5 (−89.6% anchoveta, −99.9% sardine, 0% jack mackerel by the non-identification convention of Section). Q^0_s is t 2024 Cuota Global de Captura. Artisanal fractions follow the pre-reform statutory schedule and the 2025 reform. The Los Ríos–Los Lag jack mackerel quota is reported as a single block by SERNAPESCA; the row applies the Los Lagos artisanal share (15%) as a conservat lower bound on the true sub-block share, which would rise toward 30% to the extent that Los Ríos contributes to the aggregate quo The posterior-propagated version of this table, with cross-model IQR and within-posterior 90% CI, is reported in Online Appendix H

Three readings of Table 8 structure the policy interpretation. First, the redistribution *raises* the absolute artisanal landing under projected SSP5-8.5 end-of-century stress by approximately 122 kt at the Centro-Sur aggregate, almost entirely through the 20-percentage-point increase in the artisanal jack mackerel share in Valparaíso–Los Ríos rather than through anchoveta and sardine, whose projected productivity collapse makes the 12-percentage-point increase in the artisanal share mechanically small. Second, the policy logic of the redistribution—designed to protect the artisanal sector—in fact *deepens* the artisanal exposure to anchoveta and sardine biomass collapse under climate stress: the same sector now holds 90% rather than 78% of a TAC projected to fall to floor levels. The net welfare effect therefore depends on whether the jack mackerel access expansion outweighs the deepened anchoveta-sardine exposure. Third, the counterfactual is a de jure upper bound on the post-cession allocation. Under the pre-2025 regime the industrial sector ceded 85–99% of its assigned anchoveta and sardine TACs to artisanal cessionaries via the cession channel (Online Appendix H.3), so the 12-percentage-point rise in the artisanal de jure share translates into a much smaller change in realised landings; the operationally meaningful change is the contraction of the cession volume that industrial titulares had been intermediating for fee. The reform is therefore primarily *redistributive* (transferring rent from industrial titulares to artisanal armadores) rather than *allocative* in anchoveta and sardine; for jack mackerel, historical cessions are an order of magnitude smaller and the de jure expansion is operationally meaningful.

Conclusions

This paper identifies the structural climate elasticities of stock productivity in the Chilean small pelagic fishery and combines them with a vessel-level effort equation to compute long-run comparative statics on fleet-level fishing effort under projected climate paths. We contribute in three ways. First, we couple a Bayesian state-space model with negative binomial trip equations estimated separately by sector, showing that the direction of the distributional asymmetry across fleets is governed by the interaction between portfolio composition and the limited cross-sector transferability of the prevailing allocation, extending the portfolio-and-access mechanism of Kasperski and Holland (2013), Cline et al. (2017), and Oken et al. (2021) to a climate-shock setting. Second, we identify $(\rho_s^{SST}, \rho_s^{CHL})$ sharply for the two coastal-upwelling stocks and document non-identification for jack mackerel at the Centro-Sur scale. Third, the variance decomposition shows that once the stock-dynamics steady-state collapse saturates the artisanal portfolio, the within-model component accounts for essentially all variance in projected fleet-level effort, so the policy-relevant uncertainty lies in the structural posterior rather than in CMIP6 disagreement.

The projected comparative statics indicate that the long-run intrinsic productivity of the coastal-upwelling pair—anchoveta and sardine—declines sharply under all CMIP6 scenarios considered. Sardine is consistently more exposed than anchoveta, with the SSP5-8.5 end-of-century scenario driving its steady-state productivity to floor levels at the Centro-Sur scale. The Centro-Sur jack mackerel stock, by contrast, does not provide sufficient information to identify a local climate elasticity in the 2000–2024 sample, consistent with its transboundary SPRFMO management. These results suggest that climate-adaptation policy in this fishery should target the quota-allocation architecture—specifically, the binding industrial–artisanal split and the limited cross-sector transferability that locks each fleet into its historical portfolio—rather than operational margins alone.

Several extensions are natural. A bi-level Stackelberg optimisation that uses our posterior as prior would deliver full forward biomass trajectories under endogenous TACs and effort allocation, following Kasperski (2015) with explicit cost functions and an inverse-demand system. The spatial dimension of effort allocation is a second extension, connecting the multi-species model to the location-choice literature (Dupont 1993; Smith 2005; Hicks et al. 2020); migrating the annual NB trip equation to a daily discrete-choice specification with environmentally informed SDMs as proxies for local availability would capture daily and within-port heterogeneity, following Quezada-Escalona et al. (2026). Finally, incorporating risk preferences and production risk would provide a richer characterisation of fleet responses to projected environmental variability (Kasperski and Holland 2013).

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Data and code availability

All code required to replicate the analysis is openly available at https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest. Environmental covariates are obtained from publicly available sources (E.U. Copernicus Marine Service for the historical period; CMIP6 archive via the Earth System Grid Federation for projections), and stock-assessment biomass series from IFOP and SPRFMO annual technical reports. Vessel-level logbook data are restricted-access and were obtained from IFOP under Chile's Transparency Law (Ley 20.285).

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